

Generation Z Engagement with the AAdvantage Loyalty Program

A Reddit-Based Sentiment Analysis of American Airlines



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1. PROJECT OVERVIEW

Our client, American Airlines (AA), is one of the largest global airlines. AA serves millions of passengers annually across domestic and international routes. The company operates an extensive route network and relies on its loyalty program, AAdvantage, to retain customers and drive long-term revenue. AA serves a wide range of travelers, including business professionals, frequent flyers, occasional flyers, and people of all age demographics.

Our team was tasked with analyzing the performance and perception of American Airlines' AAdvantage loyalty program, specifically focusing on Generation Z engagement. The core problem we aimed to solve was:

Why is American Airlines experiencing lower engagement and visibility in loyalty-related conversations among Generation Z, and what strategic actions can be taken to improve both attraction and retention within this segment?

Through our large-scale sentiment analysis of over 600,000 Reddit posts and comments, we investigated how American Airlines compares to competitors like Delta and United in terms of customer sentiment, engagement levels, and key pain points related to loyalty.

Loyalty programs are a primary driver of long-term customer value in the airline industry. Our findings showed that American Airlines significantly falls behind competitors in overall conversation volume and engagement related to loyalty, suggesting a lack of relevance, especially among Gen Z customers. While the sentiment levels were comparable to competitors, the lack of engagement itself signals a deeper issue: Gen Z is not actively participating in or valuing the AAdvantage program. Gen Z represents the next generation of frequent travelers and long-term customers. If AA fails to build loyalty with this group early, it risks losing lifetime customer value to competitors. Low visibility today can translate into long-term brand weakness. Addressing this issue now allows AA to reposition its loyalty strategy before disengagement becomes permanent.

2. WHERE WE STARTED

Our original project plan centered on analyzing American Airlines' proprietary AAdvantage data. To prepare, the team built a complete supervised machine learning pipeline on a publicly available Kaggle airline passenger satisfaction dataset: model-agnostic, transferable, and ready to deploy on AA's internal data once delivered. In parallel, we ran a small Reddit sentiment pilot of roughly 5,600 comments scored with VADER, intended as a supplementary qualitative layer to give the ML findings real customer voice.

The AA data was not delivered within our project timeline. The pivot that followed is detailed in section 4.

Going in, we expected Gen Z sentiment toward flying, and toward AAdvantage specifically, to lean negative. Our literature review supported this hypothesis. Gen Z is the most price-sensitive cohort, with 55% willing to switch brands for a cheaper or better deal (PayPal). They are also the most willing to abandon brands that feel inauthentic, with 32% leaving over "forced" or "cringey" communication (SKIM Group). Gen Z is significantly less likely to enroll in airline loyalty programs than older cohorts, at 65% versus 89% of Boomers, with 61% citing the lack of single-carrier travel as the primary barrier (OAG). We assumed AA would track competitors closely on raw sentiment scores but lag on engagement and visibility, particularly in loyalty-focused conversation.

Several constraints shaped what we could and could not test: **No access to AA-internal data.** We could not see actual AAdvantage enrollment, redemption, or churn metrics. All conclusions about loyalty performance had to be inferred from public conversation rather than transactional behavior. **No verifiable user demographics on Reddit.** The platform does not provide age, location, or income. Our Gen Z identification relied entirely on linguistic and behavioral proxies, meaning our "Gen Z" sample is best understood as posts that *read* as Gen Z, not posts confirmed to be authored by Gen Z users. **Reddit API restrictions.** Reddit's November 2025 Responsible Builder Policy closed standard self-service API access, forcing us to work through a third-party archive for data collection at scale. **Single platform.** We did not pull from X, TikTok, or Instagram. Findings reflect Reddit's particular user base and discussion norms, which skew toward more deliberative, text-heavy discourse than the platforms where Gen Z is most active. **Theoretical framing grounded in secondary research.** Our Gen Z behavioral model rests on published linguistic and consumer research rather than primary survey work, which was outside our scope.

3. ANALYTICAL JOURNEY

Reddit restricted self-service API access through its November 2025 Responsible Builder Policy, so we turned to Arctic Shift, a community-maintained public archive that provides the same post and comment data as Reddit's official API (Heitmann). Our scraping pipeline used date-based sliding window pagination to pull records back to January 2022, with automatic partial saves to prevent data loss on connection failures.

We built three datasets, each answering a different layer of the research question. Dataset 1 captured loyalty program sentiment: 245,345 posts from seven subreddits (r/americanairlines, r/delta, r/unitedairlines, r/awardtravel, r/churning, r/frequentflyers, and r/creditcards), filtered to records mentioning loyalty-specific keywords like AAdvantage, SkyMiles, or MileagePlus. Dataset 2 captured Gen Z travel experience: 408,816 posts from three subreddits (r/Flights, r/solotravel, and r/TravelHacks), filtered through a three-layer keyword strategy using 26 Gen Z language indicators, experience and behavior keywords, and airline-specific terms. Dataset 3 was a direct three-airline comparison: over 316,000 posts from r/americanairlines, r/delta, and r/unitedairlines, run through the same Gen Z proxy filter as Dataset 2 so findings were directly comparable. In total, the pipeline produced over 627,000 scored records.

Cleaning required multiple rounds of iteration. Reddit text contains moderator messages, bot replies, and platform boilerplate that contaminated early outputs. We built a four-category stopword architecture to isolate real signal: AIRLINE_STOPS, REDDIT_STOPS, REDDIT_NOISE, and FILLER_STOPS. NaN values required a regex-based filter because they appeared inside n-gram strings rather than as standalone entries. We also implemented word boundary matching on our Gen Z keyword filter after discovering that substring matching produced a 93% false positive rate, with terms like "mid" matching "midtown." After the fix, the false positive rate dropped to 11-47% depending on the keyword.

We generated word clouds and frequency tables at the unigram, bigram, and trigram level for each dataset, broken down by airline, sentiment label, and year. The most important early signal came from the volume breakdown: in Dataset 1, AA generated 33,995 loyalty-specific records compared to 104,120 for United and 93,126 for Delta. AAdvantage was largely absent from the loyalty conversation. When we moved to Dataset 2, the picture shifted. Gen Z was highly active in general travel subreddits at

comparable rates across all three airlines. The gap was specific to loyalty programs, not airlines. This became one of the most important distinctions in the project: Gen Z was not ignoring AA, they were ignoring AAdvantage.

All sentiment scoring used the Twitter RoBERTa model (Camacho-Collados et al.), a transformer trained on 58 million social media posts that classifies text as positive, neutral, or negative. Arriving at RoBERTa was a process of iteration. We started with VADER, a rule-based lexicon tool that underestimated negative sentiment because it scores words in isolation rather than in context. We upgraded to DistilBERT, which understood sentence-level context but only produced two classes and was trained on movie reviews rather than social media. Twitter RoBERTa resolved both limitations. Each upgrade was driven by a specific shortcoming observed in the prior model's outputs, and this progression gave us increasing confidence that our final scores reflected real patterns rather than tool artifacts.

Beyond sentiment, we used CountVectorizer for n-gram tokenization and LDA topic modeling to identify recurring themes, all broken down by airline, year, and sentiment label. Analysis was executed across three Jupyter notebooks in Google Colab with consistent methodology.

In parallel, our team built a complete supervised ML pipeline during the first half of the semester: K-Means clustering, XGBoost and LightGBM with SHAP, and logistic regression, developed on a Kaggle proxy dataset. We delivered this to American's team as a tested, transferable framework ready to run on AA's proprietary data whenever it becomes available.

4. ONE DECISION THAT CHANGED THE PROJECT

Midway through the semester, we pivoted the core analytical foundation from supervised machine learning to NLP and sentiment analysis. The Reddit sentiment work, originally a supplementary qualitative layer, became the entire backbone of the project.

The original plan centered on analyzing AA's proprietary AAdvantage customer data using the ML pipeline we had built. Despite repeated outreach through our faculty advisor and our contact at AA, the internal data team was unable to deliver the dataset within the project timeline. At that point, we had already completed an early-stage Reddit sentiment analysis using roughly 5,600 comments scored with VADER. It was a

side piece designed to add customer voice to the ML findings, but it was also the one part of the project that relied on data we controlled. We made the decision to scale it into the primary analysis.

Scaling the sentiment analysis from a supplement to the full foundation required rebuilding nearly everything. We went from one subreddit to scraping across fifteen, from 5,600 comments to over 627,000 scored records across three purpose-built datasets. The sentiment model was upgraded twice. We developed the Gen Z behavioral proxy methodology, built a four-layer stopword architecture, a word boundary regex system, and a competitive comparison pipeline putting AA, Delta, and United side by side with identical methodology.

The resulting analysis revealed something proprietary data might not have shown. AA's challenge with Gen Z is a perception and visibility problem, not an operations or product problem. AAdvantage has the lowest positive sentiment rate (10.1% vs. 13.8% for Delta and 12.8% for United), yet Gen Z discusses all three airlines at comparable volumes in travel communities. The gap is specific to the loyalty program.

We are confident this pivot was correct because the findings are actionable within the domain our client controls: perception and visibility are marketing problems with marketing solutions. However, if AA had provided enrollment data segmented by age, we could have validated whether Gen Z's absence from loyalty conversations corresponds to actual enrollment behavior, or whether they enroll at reasonable rates but simply do not discuss it online. Similarly, redemption and earn-rate data by demographic would let us test whether the negative sentiment around earning miles (48.9% of negative comments) reflects a structural disadvantage or a perception gap. The ML pipeline we delivered to AA is designed to answer exactly these questions once the data is available.

5. KEY FINDINGS AND BUSINESS INSIGHTS

The most consequential finding of this project is that American Airlines does not have a product problem - it has a perception and visibility problem. Across all three datasets, comprising more than 600,000 Reddit posts, AAdvantage appeared in only 33,447 posts compared to 93,126 for Delta's SkyMiles and 104,120 for United's MileagePlus. This represents a 65-68% mindshare gap that cannot be explained by data methodology, since all three datasets used identical sentiment models, date ranges, and

filtering logic. The business implication is direct: Generation Z is not discussing, comparing, or engaging with AAdvantage in the online spaces where they research travel decisions. When loyalty programs come up in conversation, American Airlines is conspicuously absent.

A second major finding is that loyalty programs are not part of Gen Z's travel vocabulary. Analysis of 289,500 posts from general travel communities - filtered specifically for Gen Z language patterns - revealed that loyalty programs ranked last among all complaint themes, appearing in only 1.3-3.7% of negative posts. By contrast, flight delays and cancellations dominated at 21-48%, followed closely by booking and app issues at 24-46%. Gen Z's frustrations center on immediate, tangible pain points: cost, convenience, and reliability. This finding matters because AAdvantage is architected around long-term mile accumulation and elite status - a model designed for frequent business travelers, not for a generation that travels episodically, prioritizes price transparency, and entered adulthood during economic uncertainty carrying significant student debt.

The project's most paradoxical finding emerged from the head-to-head competitive analysis of Dataset 3. American Airlines outperforms Delta and United on every major complaint category measured: flight delays and cancellations (AA at 42.2% vs. Delta at 51.5% and United at 51.4%), customer service (28.5% vs. 38.4% and 39.9%), booking and app experience (33.8% vs. 46.3% and 49.4%), pricing and fees (15.3% vs. 27.9% and 29.2%), and baggage handling (17.0% vs. 23.0% and 26.0%). AA wins on nine out of nine measured categories, yet Gen Z does not perceive it as the better airline. Brand awareness is not the issue - Gen Z discusses all three airlines at roughly equal volume in their dedicated subreddits. The problem is that AA's superior operational performance is invisible. Gen Z experiences it when they fly American, but they do not attribute it to the brand, and they certainly do not connect it to AAdvantage.

A fourth insight addresses what Gen Z actually values when travel goes well. Analysis of r/solotravel - the only dataset community with roughly balanced sentiment - showed that positive travel experiences for Gen Z center on flexibility and independence, value and pricing transparency, and the journey itself as a source of personal enrichment. Complaint rates for seats, staff, and loyalty programs in this community were the lowest across all subreddits analyzed. This suggests that AAdvantage's emphasis on upgrade benefits and elite status is structurally misaligned with how Gen Z defines a

good travel experience. To engage this demographic, rewards must feel immediate and flexible - fee waivers, refundability, date changes - not abstract miles that take years to convert into meaningful value.

Finally, the data identified a detectable and actionable "switching segment." Dataset 2 isolated approximately 925 posts combining Gen Z language patterns, a named airline, and explicit switching intent. These posts had the most negative average sentiment score of any segment (-0.739) and almost no positive content. Critically, even within this segment - travelers at the moment of maximum frustration - loyalty programs were rarely mentioned. This means AAdvantage is failing at the precise moment it should intervene. The switching segment is detectable in real time using the sentiment pipeline delivered as part of this project, creating a direct opportunity for American Airlines to intercept at-risk Gen Z customers with immediate service recovery before their loyalty migrates permanently to a competitor.

6.FINAL DELIVERABLE

Our team delivered a comprehensive analytical and strategic package designed to give American Airlines both immediate insight and long-term infrastructure for understanding and engaging Generation Z. The deliverable consists of three interdependent components: three validated datasets, a production-ready machine learning framework, and a prioritized strategic roadmap with measurable outcomes.

The analytical foundation is three original Reddit datasets totaling over 600,000 posts, each designed to answer a distinct research question. Dataset 1 (245,345 posts from loyalty-focused communities) established the competitive visibility gap between AAdvantage and its rivals. Dataset 2 (289,500 posts from general travel communities, filtered for Gen Z language) revealed what this demographic actually cares about when they fly. Dataset 3 (316,450 posts from airline-specific subreddits) provided the head-to-head competitive comparison showing AA's operational superiority and the corresponding perception failure. All three datasets used the Twitter RoBERTa sentiment model for cross-dataset consistency and were delivered with full documentation covering data collection, cleaning methodology, and analytical logic, making the analysis fully replicable on a quarterly basis.

The second component is a validated supervised machine learning framework

that American Airlines can deploy internally on its own customer data. The framework includes K-Means clustering for customer segmentation, XGBoost and LightGBM for loyalty prediction with SHAP-based feature importance analysis, and logistic regression for modeling the relationship between service factors and loyalty outcomes. It was developed and tested on a publicly available Kaggle airline passenger satisfaction dataset, and the architecture is model-agnostic, meaning it can be applied directly to AA's internal CRM, booking systems, and AAdvantage transaction data without requiring modification to the core pipeline. Once deployed on proprietary data, the framework enables automated customer segmentation, identification of the specific service levers that predict enrollment and retention, and continuous tracking of loyalty program engagement trends.

The third component is a ten-recommendation strategic roadmap organized by dataset origin and implementation timeline. The recommendations span four phases: quick wins in the first three months (deploying sentiment monitoring, simplifying AAdvantage communications, and piloting real-time complaint intercept); program redesign in months three through six (introducing fee-based immediate rewards, adding flexible redemption options, and partnering with Gen Z travel creators); strategic repositioning in months six through twelve (launching proof-based marketing that translates AA's operational advantages into authentic content on platforms Gen Z uses); and continuous optimization beyond twelve months (scaling successful pilots and monitoring Gen Z engagement metrics including share of voice, new member acquisition, and sentiment score trends). Each phase includes explicit success metrics tied directly to the dataset findings, ensuring that progress can be measured against a consistent baseline.

7. CHALLENGES AND PROBLEM SOLVING

One of the biggest lessons from this project was that real analytics work does not move in a straight line. Our team had to work through technical, data, and coordination challenges that forced us to make pivots and adjustments throughout the project. Instead of treating these obstacles as setbacks, we used them to refine our approach, improve the quality of our analysis, and make our final recommendations more grounded.

Our biggest technical challenge was working with a massive volume of

unstructured Reddit data. Unlike a clean internal customer dataset, these posts were messy, informal, and inconsistent. Many posts used slang, abbreviations, sarcasm, or airline jargon that was difficult for our sentiment model to interpret perfectly on the first attempt. We had to be careful not to treat every model output as a perfect reflection of customer opinion due to our analysis relying heavily on natural language processing. Another challenge was making sure our sentiment analysis results were useful for American Airlines to make a business decision. The model could classify posts as positive, negative, or neutral, but those classifications alone could not explain why customers felt a certain way. We realized that simply reporting sentiment percentages was not enough, but instead we needed to connect the model results to loyalty-related themes such as earning miles, status, customer service, delays, and program value.

We addressed this by moving beyond basic sentiment counts and pairing sentiment results with theme analysis. Doing so allowed us to better comprehend what customers were actually saying instead of simply classifying them as positive, negative, or neutral. This shift made our findings more actionable for American Airlines.

The most significant data challenge, the loss of AA's proprietary data and the resulting pivot to Reddit sentiment analysis, is detailed in Section 4. Beyond that pivot, the ongoing data challenge was working with a platform that offers no verifiable user demographics. Reddit does not provide age, location, or income, so every claim we make about Gen Z behavior rests on linguistic and behavioral proxies rather than confirmed identity. We addressed this by grounding our 26-keyword proxy filter in published sociolinguistic research and by validating the filter's output against known Gen Z behavioral patterns in the literature. We were transparent about this limitation throughout the report rather than treating the proxy as a confirmed demographic label.

Our biggest team challenge was maintaining coordination after the mid-semester pivot. With the project's scope and methodology changing significantly, we had to reset roles and ownership quickly while still preserving the ML pipeline as a separate deliverable. We addressed this by formalizing dataset-level ownership, where each of the three Reddit datasets had a clear primary owner, and by treating the ML pipeline as a parallel workstream maintained independently. This structure eliminated coordination bottlenecks and let the team work in parallel through the second half of the semester. The key lesson was that when a project's direction changes under deadline pressure, the

team structure has to change with it or the work stalls.

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9. APPENDIX

Detailed Complaint Rate Comparison by Airline

The following complaint rates (percentage of negative posts mentioning each issue) were derived from Dataset 3's head-to-head analysis of 316,450 Gen Z-filtered posts across r/americanairlines, r/delta, and r/unitedairlines:

Complaint Theme	AA	Delta	United	Gap
Flight Delays & Cancellations	42.2%	51.5%	51.4%	9 pts
Customer Service	28.5%	38.4%	39.9%	10 pts
Booking & App	33.8%	46.3%	49.4%	13 pts
Pricing & Fees	15.3%	27.9%	29.2%	13 pts
Refunds & Compensation	23.7%	29.1%	29.1%	5 pts
Baggage	17.0%	23.0%	26.0%	7 pts
Seats & Comfort	18.6%	22.0%	21.6%	3 pts
Staff & Crew	13.1%	17.1%	16.8%	4 pts

Gen Z Complaint Theme Prevalence Across Travel Communities

From Dataset 2 (289,500 Gen Z-filtered posts), complaint theme prevalence ranged as follows across r/Flights, r/solotravel, and r/TravelHacks:

Complaint Theme	Prevalence Range Across Communities
Flight Delays & Cancellations	21%–48%
Booking & App Issues	24%–46%
Pricing & Fees	23%–30%
Customer Service	17%–31%
Refunds & Compensation	4%–39%
Loyalty Programs	1.3%–3.7%

Data Summary

Dataset	# of Posts	Sources	Focus
Dataset 1	245,345	r/awardtravel, r/churning, r/CreditCards	Loyalty program visibility and competitive share of voice
Dataset 2	289,500	r/Flights, r/solotravel, r/TravelHacks	Gen Z travel priorities and frustration themes
Dataset 3	316,450	r/americanairlines, r/delta, r/unitedairlines	Head-to-head airline performance and perception gap

Limitation and Methodological Considerations

This analysis is grounded in publicly available Reddit data, which represents a specific subset of Gen Z travelers: those who are active in online communities, willing to share experiences publicly, and engaged enough to write detailed posts. This population skews more opinionated and more digitally native than the average Gen Z traveler. However, these are precisely the travelers who influence others at scale through Reddit discussions, TikTok videos, and YouTube reviews. The machine learning framework was validated on publicly available Kaggle data rather than American Airlines' proprietary customer data; final performance will depend on the quality and completeness of AA's internal data systems. All sentiment analysis carries inherent limitations - sarcasm, context-dependent meaning, and cultural nuance can mislead even advanced transformer models such as RoBERTa. These limitations were mitigated through manual validation of sample posts, but some classification errors are inevitable at scale. Despite these constraints, the convergence of findings across three independent datasets, multiple subreddits, and distinct analytical approaches supports confidence in the core conclusions.